01 | FOUNDATIONS / 1D KINEMATICS

Dynamics = kinematics (describes motion) + kinetics (forces causing change). **Statics**: $a = 0$ (at rest or constant velocity). **Particle**: mass but negligible size. **Rigid body**: negligible deformation. **Inertial frame**: nonaccelerating reference. **Space** locates; **time** measures succession; **mass** measures inertia.

Displacement vs distance

$\Delta s = s_2 - s_1$ is signed displacement; distance = sum of absolute path lengths. $\Delta t = t_2 - t_1$; s_{path} is total distance.

$$v_{\text{avg}} = \frac{\Delta s}{\Delta t} \qquad \bar{v}_{\text{speed}} = \frac{s_{\text{path}}}{\Delta t}$$

$$v = \frac{ds}{dt} = \dot{s} \qquad a = \frac{dv}{dt} = \ddot{s}$$

v: signed velocity [m/s]; a: signed acceleration [m/s^2]; speed $= |v|$. a and v opposite signs means slowing; $v = 0$ need not mean $a = 0$.

$$a = \frac{dv}{dt} = v\frac{dv}{ds} \;\Rightarrow\; a\,ds = v\,dv$$

Constant acceleration ONLY

$$v = v_0 + a_c t$$

$$s = s_0 + v_0 t + \frac{1}{2}a_c t^2$$

$$v^2 = v_0^2 + 2a_c(s - s_0)$$

$$s - s_0 = \frac{v_0 + v}{2}t$$

s_0, v_0: initial position/velocity at $t = 0$; a_c: constant acceleration. From $a_c = dv/dt$, $v = ds/dt$ and $v\,dv = a_c\,ds$.

When acceleration varies

$a = a(t)$: $v = v_0 + \int a\,dt$, then $s = s_0 + \int v\,dt$. $a = a(s)$: $\frac{1}{2}(v^2 - v_0^2) = \int_{s_0}^{s} a\,ds$. $a = a(v)$: $dt = dv/a$; integrate for time. $v = v(s)$: $a = v\,dv/ds$, $dt = ds/v$. **PROCESS:** Given $a(t)$: integrate in time. Given $a(s)$: use $v\,dv = a\,ds$. Given $v(t)$: integrate to position, differentiate to acceleration. Apply initial conditions for constants.

Graph interpretation

Slope $s - t \Rightarrow v$; slope $v - t \Rightarrow a$; signed area $v - t \Rightarrow \Delta s$; signed area $a - t \Rightarrow \Delta v$. For *distance*, find where $v = 0$, split time intervals and sum $|\Delta s_i|$. Multi-stage motion: final state of one stage = initial of next.

02 | CURVILINEAR / CARTESIAN

$\mathbf{r}$: position vector. $\Delta\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$; $\mathbf{v} = d\mathbf{r}/dt$ tangent to path; $\mathbf{a} = d\mathbf{v}/dt$ is not generally tangent. Averages: $\mathbf{v}_{\text{avg}} = \Delta\mathbf{r}/\Delta t$, $\mathbf{a}_{\text{avg}} = \Delta\mathbf{v}/\Delta t$. Speed $v = |\mathbf{v}| = ds/dt$.

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j}, \quad |\mathbf{r}| = \sqrt{x^2 + y^2}$$

$$\mathbf{v} = \dot{x}\mathbf{i} + \dot{y}\mathbf{j}$$

$$\mathbf{a} = \ddot{x}\mathbf{i} + \ddot{y}\mathbf{j}$$

If $a_x = a_x(x)$, then $a_x = v_x\,dv_x/dx$ (similarly $a_y = v_y\,dv_y/dy$). $\mathbf{i}, \mathbf{j}$ fixed unit vectors; $\dot{x} = v_x, \dot{y} = v_y$; $\ddot{x} = a_x, \ddot{y} = a_y$. $|\mathbf{v}| = \sqrt{v_x^2 + v_y^2}$; $|\mathbf{a}| = \sqrt{a_x^2 + a_y^2}$; direction $\tan\phi = v_y/v_x$ (check quadrant); $\mathbf{u}_v = \mathbf{v}/|\mathbf{v}|$.

Projectile (no drag; +y upward)

$$a_x = 0, \qquad a_y = -g, \qquad g = 9.81\ \text{m/s}^2$$

$$x = x_0 + (u\cos\beta)t$$

$$y = y_0 + (u\sin\beta)t - \frac{1}{2}gt^2$$

$$v_x = u\cos\beta, \qquad v_y = u\sin\beta - gt$$

u: launch speed; β: launch angle. Top: $v_y = 0$ but $a_y = -g$. **PROCESS:** Find $t \geq 0$ from vertical equation, then horizontal range; use components for impact velocity.

03 | NORMAL–TANGENTIAL (n-t)

Use when path/curvature known. $\mathbf{u}_t$: tangent along velocity; $\mathbf{u}_n$: inward toward centre of curvature. ρ: radius of curvature; $v = |\mathbf{v}| \geq 0$.

$$\mathbf{v} = v\mathbf{u}_t, \quad \mathbf{a} = a_t\mathbf{u}_t + a_n\mathbf{u}_n$$

$$a_t = \frac{dv}{dt} = v\frac{dv}{ds} \qquad a_n = \frac{v^2}{\rho}$$

a_t: change of speed (signed); a_n: change of direction (inward, nonnegative); $|\mathbf{a}| = \sqrt{a_t^2 + a_n^2}$. Straight path $\rho \to \infty \Rightarrow a_n = 0$; circle $\rho = R$. Constant speed along curve still gives $a_n \neq 0$.

$$\rho = \frac{[1 + (dy/dx)^2]^{3/2}}{|d^2y/dx^2|}$$

$y = f(x)$; tangent angle satisfies $\tan\phi = dy/dx$. Rotating tangent: $d\mathbf{u}_t/dt = (v/\rho)\mathbf{u}_n$, from $ds = \rho\,d\phi$.

Circular motion (constant radius R)

$$v = R|\omega| \qquad a_n = R\omega^2 = \frac{v^2}{R}$$

$\omega = d\theta/dt$ [rad/s]; $\alpha = d\omega/dt$ [rad/s^2]; $a_t = dv/dt$, magnitude $R|\alpha|$. **PROCESS:** Draw tangent and inward normal → find ρ, v, a_t, a_n → resolve forces along t,n.

04 | POLAR: ROTATING ARM / SLOT

r: distance from fixed origin O, θ: CCW angle; $\mathbf{u}_r$: radially outward, $\mathbf{u}_\theta$: 90° CCW from $\mathbf{u}_r$; $\omega = \dot{\theta}, \alpha = \ddot{\theta}$. Position $\mathbf{r} = r\mathbf{u}_r$.

$$\dot{\mathbf{u}}_r = \omega\mathbf{u}_\theta, \quad \dot{\mathbf{u}}_\theta = -\omega\mathbf{u}_r$$

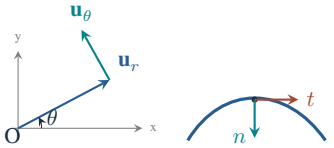
$$\mathbf{v} = \dot{r}\mathbf{u}_r + r\omega\mathbf{u}_\theta$$

$$\mathbf{a} = (\ddot{r} - r\omega^2)\mathbf{u}_r + (r\alpha + 2\dot{r}\omega)\mathbf{u}_\theta$$

$$v_r = \dot{r}, \; v_\theta = r\omega, \; a_r = \ddot{r} - r\omega^2$$

$$a_\theta = r\alpha + 2\dot{r}\omega$$

$\ddot{r}$: radial-rate change; $-r\omega^2$: inward centripetal term; $r\alpha$: angular-acceleration term; $2\dot{r}\omega$: Coriolis term (rotating-coordinate effect, also encountered with Earth rotation). $|\mathbf{v}| = \sqrt{v_r^2 + v_\theta^2}$, $|\mathbf{a}| = \sqrt{a_r^2 + a_\theta^2}$. **Fixed r:** $\dot{r} = \ddot{r} = 0 \Rightarrow a_r = -r\omega^2$, $a_\theta = r\alpha$. **Constant angular speed:** $\alpha = 0$, but $2\dot{r}\omega$ may remain. For $r = f(\theta)$, $\dot{r} = (dr/d\theta)\omega$, $\ddot{r} = (d^2r/d\theta^2)\omega^2 + (dr/d\theta)\alpha$. **PROCESS:** Obtain $r, \dot{r}, \ddot{r}, \omega, \alpha$ from geometry/time; calculate polar components, then magnitudes and forces.



05 | RELATIVE TRANSLATION

Axes attached to B translate but *do not rotate*. A/B = position, velocity or acceleration of A seen from B.

$$\mathbf{r}_A = \mathbf{r}_B + \mathbf{r}_{A/B}$$

$$\mathbf{v}_A = \mathbf{v}_B + \mathbf{v}_{A/B}$$

$$\mathbf{a}_A = \mathbf{a}_B + \mathbf{a}_{A/B}$$

$\mathbf{v}_{A/B} = \mathbf{v}_A - \mathbf{v}_B$; split into fixed x,y components; never subtract speed magnitudes as though they were vectors.

06 | PLANE KINETICS / FORCES

Newton's 2nd law: unbalanced resultant causes acceleration in its direction. Use inertial axes and constant mass.

$$\sum \mathbf{F} = m\mathbf{a} = \frac{d(m\mathbf{v})}{dt}$$

m [kg]; $\mathbf{F}$ [N]; $\mathbf{a}$ [m/s^2]. For a particle system, $M = \sum_i m_i$, G = centre of mass:

$$\sum \mathbf{F}_{\text{ext}} = \sum_i m_i\mathbf{a}_i = M\mathbf{a}_G$$

D'Alembert: $\sum \mathbf{F} - m\mathbf{a} = 0$ ($-m\mathbf{a}$ is fictitious inertia term, not real FBD force).

FBD and kinetic diagram

Isolate each body/system. Draw *real* forces: $W = mg$ down, contact normal N perpendicular to surface, tension T along cord away from body, applied F, friction against slide, spring restoring force. Draw $m\mathbf{a}$ on *separate* kinetic diagram.

$$F_k = \mu_k N \qquad |F_{s,\text{static}}| \leq \mu_s N$$

μ_k, μ_s: sliding/static friction coefficients; k: spring stiffness [N/m], s: deformation; spring force magnitude ks. $N \neq mg$ in general.

Equations in appropriate axes

$$\sum F_x = ma_x, \quad \sum F_y = ma_y$$

$$\sum F_t = m\frac{dv}{dt}, \quad \sum F_n = m\frac{v^2}{\rho}$$

$$\sum F_r = m(\ddot{r} - r\omega^2)$$

$$\sum F_\theta = m(r\alpha + 2\dot{r}\omega)$$

For plane n-t $\sum F_b = 0$ (binormal). **Centripetal force** = net *inward component* of existing forces, not an extra FBD force. If constrained to no acceleration in a direction, sum of forces there is zero.

Inclines / contact

Along incline β: $mg\sin\beta$ down slope. Into surface: $mg\cos\beta$. Normal $N = mg\cos\beta$ only if no other normal components/acceleration. At threshold of losing contact $N = 0$.

07 | CABLE / SYSTEM CONSTRAINTS

With inextensible rope, total length is constant. For a typical n-span layout: n: number of moving spans supporting body B; C: fixed-length segments. Choose *same coordinate sign convention*.

$$s_A + ns_B = C$$

$$\Delta s_A = -n\Delta s_B, \quad v_A = -nv_B$$

$$a_A = -na_B$$

An ideal same-cable tension has magnitude T throughout; the total supporting pull on moving pulley can be nT. Movable pulley speeds are generally *not equal*. **PROCESS:** Label lengths → $L = C$ → differentiate for speeds/accelerations → use Newton or energy.

08 | METHOD / QUICK CHECKS

Instant force/acceleration: Newton + FBD. **Time/displacement**: kinematics. **Speed across positions**: work–energy. **Short impact**: impulse–momentum. **Moment around a point**: angular momentum. **PROCESS:** 1 Axes/signs; 2 FBD + kinetic diagram; 3 governing equation; 4 geometry/constraints; 5 solve; 6 units/direction. Negative result means assumed arrow was opposite.

Watch: Never assume constant acceleration for variable force, normal $= mg$ on a curve, or $\dot{r} = 0$ merely because ω constant.

Useful conversions

1 rev $= 2\pi$ rad; rpm $\times \frac{2\pi}{60} =$ rad/s. 1 Mg $= 1000$ kg; km/h $\div 3.6 =$ m/s; 1 hp $= 550$ ft lbf/s $\simeq 746$ W.

09 | WORK OF FORCES

Work U [J=N m] = scalar energy transferred by a force over displacement of its point of application. ϕ: instantaneous angle between force and displacement.

$$dU = \mathbf{F} \cdot d\mathbf{r} = F\cos\phi\, ds$$

$$U_{1\to2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r}$$

Constant F and direction: $U = F(s_2 - s_1)\cos\phi$. Varying $F_\parallel$: $U = \int_{s_1}^{s_2} F_\parallel(s)ds$, the signed area under the $F_\parallel - s$ graph. Along motion +, against motion –, perpendicular 0. Stationary fixed-support reaction does no work.

Weight, spring, sliding friction

$$U_g = mg(y_1 - y_2) = -mg\Delta y$$

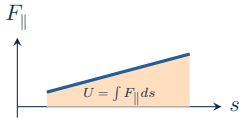
m: mass, g: gravity, y: height positive upward. Falling = positive work by weight.

$$U_s = \frac{1}{2}k(s_1^2 - s_2^2)$$

k: stiffness [N/m]; $s_i = L_i - L_0$: deformation from *natural* spring length L_0 (can be signed). Restoring spring force along deformation $F_s = -ks$.

$$U_f = -\int \mu_k N\, ds$$

μ_k: kinetic friction coefficient; N: normal reaction; ds: sliding distance. Constant N: $U_f = -\mu_k N\ell$, where ℓ is *total slip distance*.



10 | KINETIC ENERGY / WORK–ENERGY

T = kinetic energy [J]; m = mass, v = *speed* (≥ 0).

$$T = \frac{1}{2}mv^2$$

$$T_1 + \sum U_{1\to2} = T_2$$

$$\sum U_{1\to2} = \frac{1}{2}m(v_2^2 - v_1^2)$$

For a system, sum $\frac{1}{2}m_i v_i^2$ for *each* moving mass, and all external work. Count each transfer once. Useful for variable forces and speed from displacement without time. **PROCESS:** Choose states 1 and 2 → FBD and displacement → signed work of each force → energy equation → speed/position; if time needed, integrate $dt = ds/v(s)$.

11 | POWER / EFFICIENCY

$$P = \frac{dU}{dt} = \mathbf{F} \cdot \mathbf{v} = Fv\cos\phi$$

P power [W=J/s]; $\mathbf{v}$ velocity at *application point* of $\mathbf{F}$; ϕ force–velocity angle. If collinear and $v \neq 0$, $F = P/v$.

$$\eta = \frac{P_{\text{useful,out}}}{P_{\text{in}}} = \frac{E_{\text{useful,out}}}{E_{\text{in}}}$$

η: efficiency $0 \leq \eta \leq 1$; inputs, outputs measured over equal time. $P_{\text{in}} = P_{\text{out}}/\eta$. At steady velocity $\sum F_t = 0$. If power changes at an instant, use the new force $F = P/v$ and $\sum F_t = ma_t$ for acceleration.

12 | CONSERVATIVE / POTENTIAL

Conservative: work independent of path, depends on endpoints (gravity, ideal spring); $U_c = V_1 - V_2$. **Nonconservative:** path-dependent, often dissipative (friction). **Potential energy** V is due to position/configuration.

Gravity (arbitrary datum)

$$V_g = mgy = Wy$$

m: mass, $W = mg$: weight; y: vertical height from datum. Datum: $V_g = 0$; below datum: negative. Choose the same datum for both states.

Ideal spring (natural length as reference)

$$V_e = \frac{1}{2}ks^2 \geq 0$$

k spring constant [N/m]; $s = L - L_0$, deformation in extension *or* compression. Multiple springs: $V_e = \sum_i \frac{1}{2}k_i s_i^2$.

$$V = V_g + V_e \qquad E = T + V$$

V total potential; T kinetic; E mechanical energy. If a body momentarily stops at a turning point, set its *actual* speed to zero, not every unrelated body's speed.

13 | CONSERVATION OF ENERGY

Only conservative forces contribute work:

$$T_1 + V_1 = T_2 + V_2$$

With nonconservative work (U_{nc}):

$$T_1 + V_1 + U_{nc,1\to2} = T_2 + V_2$$

U_{nc}: signed total work of nonconservative forces, e.g. friction negative, motor work possibly positive. For a system, sum all bodies' T and all potentials. **Watch:** If a spring or gravity is included in potential V, do not add its work *again*.

Energy bookkeeping: repeatable process

PROCESS: 1 Select states 1 and 2 and height datum. 2 Calculate heights y, spring stretch $s = L - L_0$ from geometry. 3 Determine nonconservative work. 4 Use rope constraints for each body's speed. 5 List $T_1, V_1, U_{nc}, T_2, V_2$. 6 Solve/check physical root and direction.

For diagonal spring, length $L = \sqrt{h^2 + b^2}$ from perpendicular offsets. For spring engaged only after clearance c, calculate stretch *after* contact (piecewise).

What energy cannot directly supply

Energy often yields v but *not* instantaneous tension or support reaction. Use Newton $\sum F_n = mv^2/\rho$ after speed is known. It also does not determine *sign of velocity* from v^2 alone.

$$\Delta t = \int_{s_1}^{s_2} \frac{ds}{v(s)}$$

Time from a known speed-versus-position relation (if $v > 0$ along chosen increasing coordinate).

14 | PULLEYS + MULTIBODY ENERGY

For an inextensible rope, sum lengths of all variable straight segments. n: count of segments moving with body B; C: fixed remainder.

$$L = s_A + ns_B + C = \text{constant}$$

$$\Delta s_A = -n\Delta s_B, \quad v_A = -nv_B, \quad a_A = -na_B$$

$$T_{\text{total}} = \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2$$

Use $v_A^2 = n^2 v_B^2$ for energy, and $W_A\Delta s_A + W_B\Delta s_B$ with *consistent signs*. Ideal same-cable tension equal; support reactions at fixed supports do zero work, and ideal internal cable work cancels for the full system.

15 | ENERGY SIGN CHECK

Downward motion ⇒ V_g ↓; spring compression/stretch ⇒ V_e ↑; friction ⇒ energy lost. T, V, U [J]; P [W]; spring k [N/m]. Never assume all connected bodies have equal speed.

16 | LINEAR IMPULSE + MOMENTUM

Momentum p [kg m/s] is mass times velocity. **Impulse I** [N s] is force integrated over its duration (signed area under force–time graph).

$$\mathbf{p} = m\mathbf{v} \quad \mathbf{I} = \int_{t_1}^{t_2} \mathbf{F}\, dt$$

$$m\mathbf{v}_1 + \sum \mathbf{I}_{1\to2} = m\mathbf{v}_2$$

m mass; $\mathbf{v}_1, \mathbf{v}_2$ initial/final velocities; $\mathbf{F}$ external force; t_1, t_2 endpoints. For x component:

$$mv_{x1} + \sum \int_{t_1}^{t_2} F_x\, dt = mv_{x2}$$

Likewise y and z. Constant force: $I = F\Delta t$. Variable force: integrate. Average force $F_{\text{avg}} = I/\Delta t$. Account for normal/weight impulses if significant.

Particles, collision and explosion

$$\sum_i m_i\mathbf{v}_{i1} + \mathbf{I}_{\text{ext}} = \sum_i m_i\mathbf{v}_{i2}$$

Equal/opposite internal impulses cancel. If $I_{\text{ext,x}} \simeq 0$, then *x momentum* conserved:

$$\sum_i m_i v_{ix1} = \sum_i m_i v_{ix2}$$

$M = \sum_i m_i$; mass-centre $\mathbf{v}_G = \sum_i m_i\mathbf{v}_i/M$. If no external impulse ⇒ $\mathbf{v}_{G1} = \mathbf{v}_{G2}$. **Impulsive** force = large force in brief collision/explosion. **Nonimpulsive** = negligible impulse during that event (often weight). Momentum conserved only in directions of negligible external impulse. Collision/explosion may *not* conserve kinetic energy. **PROCESS:** Choose interval/system → initial/final velocity diagram → significant external impulses → x,y,z momentum equations → unknowns. For explosions, find centre/pre-event velocity first.

17 | ANGULAR MOMENTUM (TUTORIAL 6)

About a *fixed inertial point* O, $\mathbf{r}$ runs from O to particle; $\mathbf{v}$ particle velocity; $\mathbf{F}$ applied force.

$$\mathbf{H}_O = \mathbf{r} \times (m\mathbf{v})$$

$$\sum \mathbf{M}_O = \frac{d\mathbf{H}_O}{dt}, \quad \mathbf{M}_O = \mathbf{r} \times \mathbf{F}$$

$\mathbf{H}_O$: angular momentum [kg m²/s]; $\mathbf{M}_O$: force moment [N m]. Right-hand rule for cross products.

$$\mathbf{H}_{O1} + \int_{t_1}^{t_2} \sum \mathbf{M}_O dt = \mathbf{H}_{O2}$$

If external angular impulse about O/selected axis vanishes, angular momentum about that axis is conserved: $H_1 = H_2$.

$$H_z = mrv_\theta = mr^2\omega$$

r: distance to rotation axis; $v_\theta = r\omega$: transverse speed; ω: angular velocity. *Radial* speed does not contribute to H_z; overall $v = \sqrt{v_r^2 + v_\theta^2}$.

$$\mathbf{r} \times \mathbf{q} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ q_x & q_y & q_z \end{vmatrix}$$

PROCESS: Choose O/axis → locate moment-producing forces → apply angular momentum balance/conservation → combine with work–energy if total speed or work is sought.

18 | PRACTICE-PAPER ADDITIONS

DIRECT CENTRAL IMPACT (Q13)

$$e = \frac{v_B - v_A}{u_A - u_B}$$

approach: uA > uB
after: vA, vB

Momentum: mA uA + mB uB = mA vA + mB vB
e = 0: same final v (sticking); e = 1: elastic.
PROCESS: choose + axis, solve momentum + e.

F-t IMPULSE: signed area; triangle = bh/2; rectangle = bh; trapezoid = (a+b)h/2.
COMBINE: central H + energy; speed + F_n.
Check: FBD / signs / units / direction / no double count.